

SOFTWARE ENGINEERING LABS

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

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Section	5th Semester K Section
Project	Municipal Infrastructure Damage Reporting App

Project Overview

The Municipal Infrastructure Damage Reporting App was organized in Jira using Epics, User Stories, story-point estimation and two simulated Scrum sprints. The main workflow covers citizen damage reporting, photo upload, GPS-based location identification, report dispatch and engineer-side report management.

1. EPICs

The Lab 1 functional requirements were grouped into four major functional themes. The following Epics were created in the Jira workspace.

Epic	Description	User Stories
MIDRA-1 Damage Report Submission	Handles submission of infrastructure damage reports and supporting photographs.	MIDRA-5 — Submit Damage Report MIDRA-6 — Upload Damage Photo
MIDRA-2 Location Identification	Identifies the incident location using GPS information extracted from the uploaded photograph.	MIDRA-7 — Extract GPS Location
MIDRA-3 Report Dispatch	Routes each damage report to the appropriate municipal engineering department.	MIDRA-8 — Dispatch Damage Report
MIDRA-4 Engineer Report Management	Allows municipal engineers to view assigned reports and update their status.	MIDRA-9 — Manage Damage Report

Jira Timeline showing the project Epics

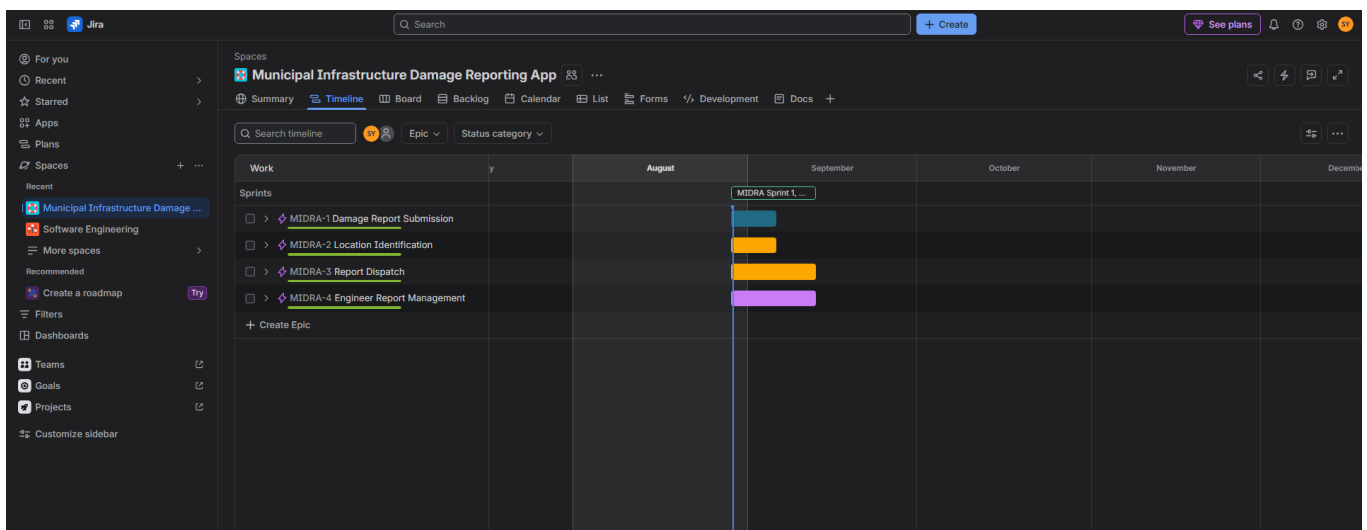


Figure 1: Four MIDRA Epics created in Jira.

2. BACKLOG AND STORY POINT ASSIGNMENTS

The backlog contains the User Stories under their respective Epics. The captured Jira view also shows the story-point estimates used during sprint planning. Sprint 2 contains the final two stories with 5 and 3 story points respectively.

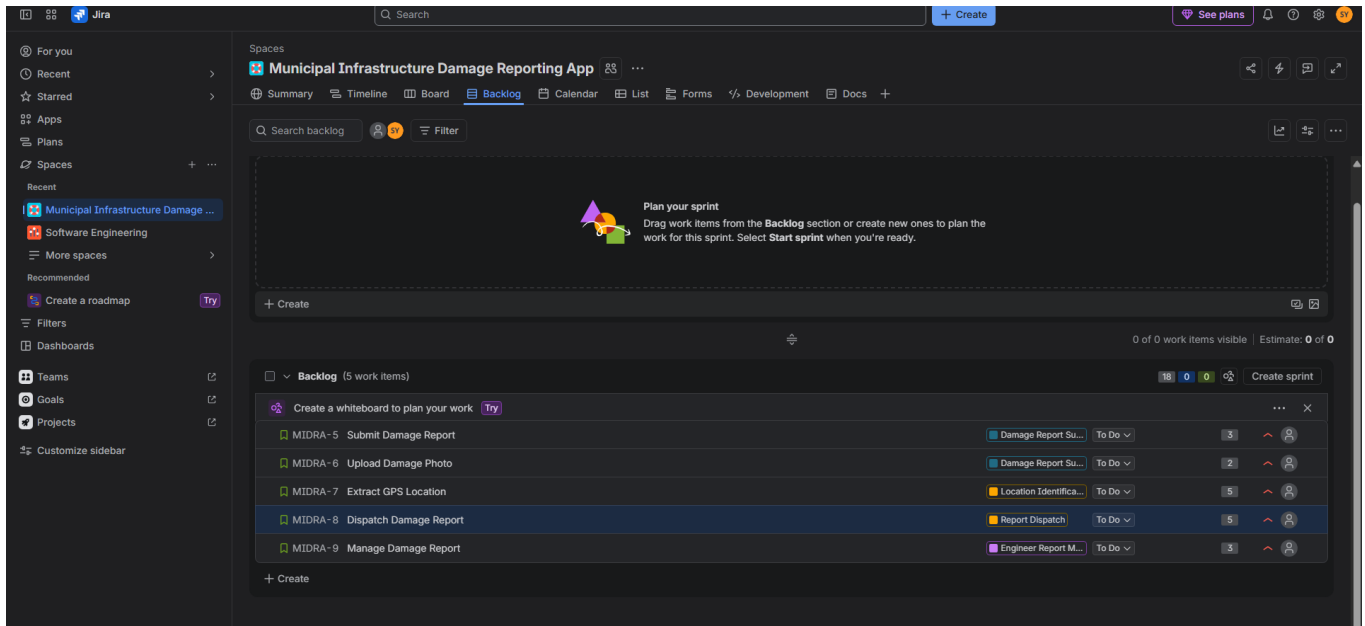


Figure 2: Jira backlog with User Stories, Epic association and story-point estimates.

3. SPRINT EXECUTION

Two sprints were completed as part of the simulation. Sprint 1 covered the initial reporting and location workflow, while Sprint 2 contained the dispatch and engineer-management stories. The Sprint 2 backlog view shows MIDRA-8 and MIDRA-9 selected as the remaining sprint work.

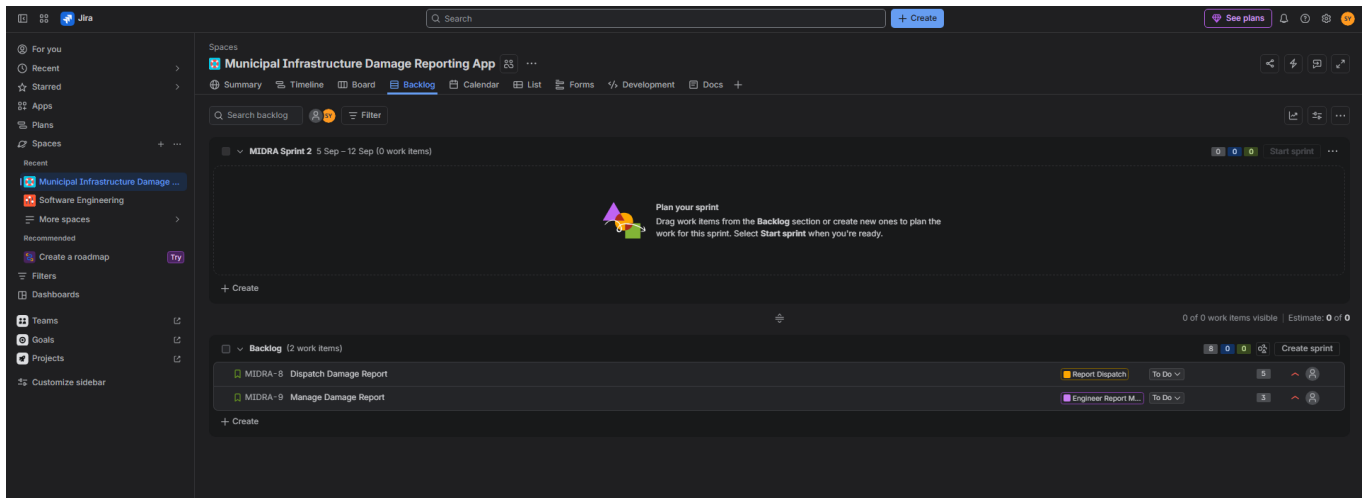


Figure 3: Sprint 2 planned work in the Jira backlog.

4. BURNDOWN CHART

The Sprint 1 Burndown Chart was generated using Story Points as the estimation field. The sprint contained 9 story points. The red line represents remaining work and the grey line represents the ideal guideline for linear progress.

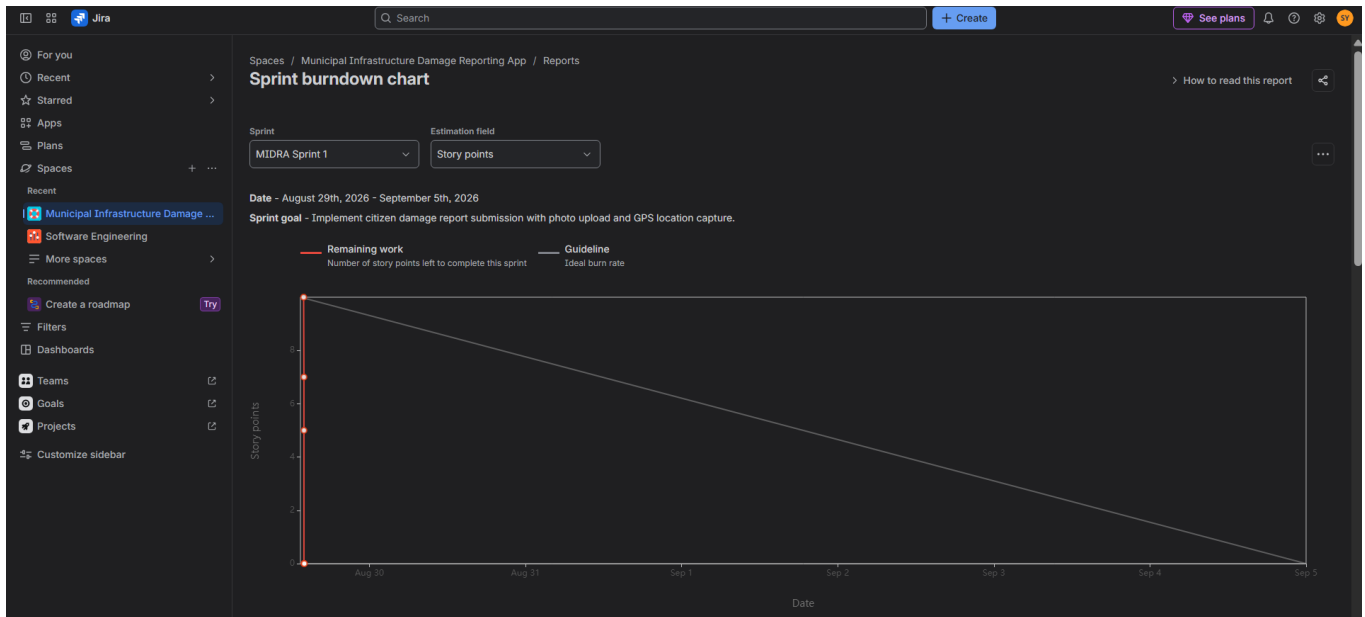


Figure 4: Sprint 1 Burndown Chart.

Observation: The chart shows a sharp reduction of remaining work to zero near the beginning of the recorded sprint period. Since this was a simulated sprint, the stories were completed in a short period. The chart therefore confirms completion of the planned workload, while also showing that the work was not spread gradually across the full sprint duration.

5. REFLECTION QUESTIONS AND ANSWERS

1. Did your estimations reflect the actual effort?

Yes, the estimates were reasonably close to the effort involved in the simulated work. The smaller tasks, such as uploading a damage photo and extracting the GPS location, were relatively straightforward. The estimates were useful when deciding which stories could be taken into each sprint.

2. Was your backlog well-prioritized?

Yes. The backlog was arranged around the main workflow of the application. Reporting the damage, attaching evidence and identifying its location were treated as important functions, followed by dispatching the report and managing it from the engineer side. This made the backlog easier to divide into sprints.

3. How did your simulated sprint align with your plan?

The simulated sprints followed the planned Scrum workflow. The selected stories were placed in the sprint and moved through To Do, In Progress and Done. Two sprints were completed, with the first sprint focusing on citizen reporting and location identification and the second on dispatch and engineer management.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart showed that the planned story points could be completed within the sprint. The quick drop to zero also reflected the fact that the sprint was simulated and the work was completed in a short period. In a real project, such a sharp drop could suggest that work was completed in large batches or that the estimates were not distributed evenly through the sprint.

Conclusion

The lab demonstrated the use of Jira for converting requirements into Epics and User Stories, estimating work using story points, planning sprints and tracking progress with a Burndown Chart. The Municipal Infrastructure Damage Reporting App was successfully organized and simulated through two completed sprints.